LESSON PLAN

HELICOPTER THEORY (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	27	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Boundary layer, laminar, transition, turbulent B/L
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Boundary layer, laminar, transition, turbulent B/L,
5. Specific Objectives	11.1.1.1 Trainee will be able to explain the properties of atmosphere, flow over bodies, boundary layer.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of IB diagrams showing boundary layer separation; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the definition of a boundary layer?
(b) How does viscosity affect flow over a body?
(c) Why is the boundary layer important for helicopter performance?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of boundary layer • Viscosity and flow separation • Laminar flow characteristics • Transition zone properties • Turbulent flow characteristics	L & Ds	PPT, IB	15 min
(b)	• Comparison of laminar and turbulent profiles • Skin friction drag mechanisms • Effects of surface roughness • Reynolds number significance • Flow separation criteria	L & Ds	PPT, PDF	20 min
(c)	• Practical implications for rotor blades • Control of boundary layer separation • Impact on lift and drag • Mitigation techniques • Real-world helicopter examples	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between laminar and turbulent boundary layers.
(b) Describe how transition occurs in a boundary layer.
(c) How does boundary layer separation affect helicopter rotor efficiency?

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of boundary layer and transition will be applied in the study of aerofoil geometrical characteristics in LP 07.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Geometrical characteristics of aerofoil, types of aerofoil, definition of terms.	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Commander, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min